Gabay: Shipping a Public RAG Product

Gabay sa bawat tanong. Cited answers on Philippine worker benefits, from official handbooks.

Reflection · Graded Mini Project 14.3 · 2026-10-11 · gabay-fint.vercel.app · github.com/jacobvillar/gabay

1 Overview

Gabay answers questions about **Philippine statutory worker benefits** (13th-month pay, leave, SSS, PhilHealth, Pag-IBIG) strictly from official government handbooks, and shows the exact source pages under every answer. It runs on **Next.js + the Vercel AI SDK** with retrieval exposed as a tool call, **Groq** (open-weight GPT-OSS 120B) for generation, and **Upstash Vector** for dense search, deployed publicly on Vercel.

2 Key decisions

Decision	Choice	Why
Corpus	DOLE handbook, SSS Citizens Charter, PhilHealth Konsulta circular, Pag-IBIG briefer (1,305 chunks)	Public-domain government works, legal to serve; a domain where base models get figures subtly wrong, so citations add real value
Chunking	Per-page extraction with PDF ligature repair, then sentence/paragraph-aware packing to ~1,000 chars, 150 overlap	Per-page parsing keeps real **page numbers** for citations; boundaries fall on sentence ends so rules stay intact
Metadata	`document_title`, `agency`, `page`, `source_url`	Readable citations that open the PDF at the cited page; `agency` also drives retrieval
Retrieval	Hybrid: dense + BM25 keyword, reciprocal rank fusion, optional per-agency boost; topK 8	Keyword catches exact terms dense misses; the boost counters the SSS-heavy corpus without ever hiding other handbooks
Generation	Groq GPT-OSS 120B; falls back to 20B on rate limits; two-search budget per question	Free tier with tool calling; the budget guarantees every reply ends in a written answer
Guardrails	Figures must appear in retrieved passages; decline off-topic; request validation; UI strips leaked LaTeX/HTML	A wrong peso figure is worse than no answer; the prompt alone proved insufficient, so the UI is a backstop
Interface	Onyx-style sidebar: New chat, Search, Documents, Recents with Rename/Delete (browser-local)	No accounts or server storage. Deliberately no admin upload, which would let anyone poison a public index

3 What it does well

Grounded, cited answers on PH-specific rules. “How is 13th-month pay computed?” triggers retrieval, answers from the DOLE handbook (1/12 of basic salary, with the correct exclusions), and each source chip opens the PDF at that page. When a figure is not in the handbooks, it says so instead of inventing one.

4 What it does badly

Retrieval balance: the SSS charter is 80% of the index and mostly procedure, so unscoped searches still lean SSS. The free tiers are the other weak spot: about 10–15 full answers a day before the smaller fallback model takes over, and it is noticeably less faithful.

5 The trickiest decision

Making retrieval **trustworthy on high-stakes questions**. Retrieval had to be a deliberate tool call rather than always-prepended context, with a tool description that makes the model reach for the handbooks on benefit questions but skip small talk, and a system prompt kept strict: answer only from retrieved passages and say plainly when they do not cover something. Balancing that faithfulness against still being helpful was the hardest part, and it surfaced again in the agency boost: a hard filter looked cleaner but let one wrong model guess hide the right handbook.

6 How I tested it

An end-to-end eval of 10 questions against production (all four agencies, Taglish, and an off-topic probe), reading every answer. It caught real defects: replies that ended empty because the model kept searching, PDF text broken 2,316 times (“bene fi t”), figures taken from model memory, an off-topic question answered, raw HTML and LaTeX in answers, an unanswerable suggested question, and a filter that hid the right handbook. Each fix was re-run against the same questions, and the seed now verifies itself with a known-answer probe (13th-month pay must return DOLE pp. 50–54).

7 What I’d do with another week

1. **Automate the eval.** Turn the 10-question run into a golden set with faithfulness and context-relevance scorers and an LLM judge calibrated on hand labels, run in CI on every prompt or retrieval change.
2. **Cross-encoder reranking** over the fused dense + keyword results, the missing second stage.
3. **Exact-match response cache** for first-turn questions (the suggested questions dominate traffic), keyed by normalized question + corpus and prompt version so a re-seed invalidates it. No semantic cache.
4. **Rebalance the corpus** by trimming the SSS charter to its benefit sections.
5. **Inline citations**: real [1]/[2] markers tied to the sources panel.

8 References

Live app gabay-fint.vercel.app · repository github.com/jacobvillar/gabay · corpus: DOLE, SSS, PhilHealth, Pag-IBIG official handbooks.

Interface design language inspired by Onyx.